

--- Page 1: Project Description & General Information ---

Project Name: GreenHorizon Super Thermal Power Plant Expansion (Stage-II)

Proponent: GreenHorizon Energy Corp. Ltd.

Location: Angul District, Odisha

Project Capacity: 2 x 660 MW Ultra-Supercritical Coal Units

Total Estimated Capital Cost: INR 4,500 Crores

Land Requirement: 420 Hectares (Private Agricultural: 310 Ha, Forest Land: 110 Ha)

Water Requirement:

The total fresh water requirement for cooling tower makeup and boiler feed is estimated at 3,500 m³/day, to be c

Power Evacuation: 400 kV D/C transmission line connected to National Grid.

--- Page 2: Baseline Environmental Data & EMP ---

Baseline Monitoring Period: October to December (Post-Monsoon Season)

Ambient Air Quality Baseline:

- PM10: 78.4 ug/m3
- PM2.5: 42.1 ug/m3
- SO2: 24.6 ug/m3
- NOx: 31.2 ug/m3

Emission Control: High-efficiency Electrostatic Precipitators (ESP) designed for 99.9% particulate collection.

Stack Height Details:

Narrative specifies a twin-flue stack of height 220 meters, while Technical Design Table 4.2 lists stack height as

Water Balance & Effluent Management (Table 4.5):

- Daily Fresh Water Intake Requirement: 5,200 m3/day (Notice: contradicts Page 1 figure of 3,500 m3/day).
- Cooling Tower Blowdown: 1,200 m3/day
- Ash Handling Slurry Water: 850 m3/day
- Domestic Water Consumption: 150 m3/day

Groundwater Conditions:

Baseline hydrological survey indicates existing shallow aquifer water tables have declined by 2.4 meters over the

--- Page 3: Land Use & Environmental Safeguards ---

Land Use Breakdown:

- Main Plant & Boiler Turbine Generator: 160 Hectares
- Ash Disposal Dykes: 110 Hectares
- Coal Storage Yard & Rail Siding: 60 Hectares
- Greenbelt Development: 90 Hectares (approx 21.4% of total area, below statutory 33% requirement)

Groundwater Impact Claim:

Section 6.4 Narrative asserts: 'The project will operate under strict zero impact guidelines and cause absolutely no impact on groundwater resources' (Notice: Contradicts Page 2 empirical baseline finding showing active aquifer drawdown).

Mandatory Studies Note:

This report incorporates Baseline environmental monitoring, Project Description, Land Use data, and an Environmental Impact Assessment. (Note: Public Hearing proceedings and Biodiversity Impact Assessment studies are pending).